

Cheng-Hao (Howard), Kao



About me

A second-year MASc student in Mechatronics engineering at the University of Waterloo. Proficient in machine learning and data analysis. Possess the analytical abilities required for research and design as well as a comprehensive grasp of engineering principles. Currently researching on multi-agent system workloads as an intern at Huawei, and working in the Structural Biomechanics Lab on ML-based injury prediction. Previously involved in the Robotics and Computer Vision Lab conducting research on multi-robot collaboration joint trajectory planning and control systems and in the Microelectromechanical Systems Lab dedicated to integration of FPCB mirror-based scanners. Seeking more professional training for further development within the field of AI, data analysis, and cognitive science.

Contact

- 👤 Born on 1999/11/30, Age 25
- ✉️ howardkao1130@gmail.com
- ☎️ +1 6476716097
- 🏠 Cheng-Hao Kao
- 🎓 Cheng-Hao Kao
- 🌐 Cheng-Hao Kao
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Languages

Chinese - Native Language
English - Professional Knowledge

Education

January
2024 -
Present

MASc in Mechanical and Mechatronics Engineering 📍 Waterloo, On, Canada
University of Waterloo
GPA: 3.90/4.0 Cumulative

Relevant Coursework: Introduction to Machine Learning, Pattern Recognition and Machine Learning, Deployment of Deep Learning Model, Computational Intelligence, Optimization Methods, Human Factors in Testing

August 2018
- June 2023

Bachelor of Engineering 📍 Toronto, On, Canada
Toronto Metropolitan University
GPA: 3.85/4.33 Cumulative

Relevant Coursework: Mechanical Design, Control of Robotic Manipulators, Real-Time Computer Control Systems, Fundamentals of Robotics, Mechatronics Systems Design, Measurements, Sensors and Instruments, Microprocessor Systems, Control Systems Digital Systems, Computer Programming Fundamentals, Electric Machines and Actuators, Electric Circuits, Applied Finite Elements, Vibrations, Machine Design, Mechanics of Machines, Solid Mechanics, Statics and Mechanics of Materials, Dynamics, Introduction to Engineering Design, Engineering Graphical Communication, Numerical Analysis, Linear Algebra, Differential Equations and Vector Calculus, Calculus II, Calculus I

August 2015
- June 2018

High School Diploma 📍 Shanghai, China
No.2 High School of East China Normal University
Average: 95.3/100

Research Experience

August 2025
- Present

Technology Planning and Cooperation Research Intern 📍 Waterloo, On, Canada
Huawei Canada

- Multi-agent system workload trend identification
- Multi-agent system system-level optimization analysis
- Multi-agent system analysis in AI for scientific research

May 2024 -
April 2024

Teaching Assistant 📍 Waterloo, On, Canada
University of Waterloo

- Teaching Circuits
- Teaching Mechanics of Deformable Solids
- Teaching Statics

June 2023 -
January
2024

Research Assistant 📍 Toronto, On, Canada
Toronto Metropolitan University RCVL Lab

- Research on multi-robot collaboration joint trajectory planning and control
- Research on micro-plastic detection via ultrasonic sensor and machine learning

Professional Skills

Deep LearningMachine Learning

Data AnalysisOptimization

Signal ProcessingRoboticsControl Systems

MicrocontrollersSimulation and Design

Engineering Mathematics

Honors and Awards

Dean's List 2018 to 2019Dean's List 2019 to 2020

Dean's List 2020 to 2021Dean's List 2021 to 2022

Mechanical Engineering Alumni Award

Mechanical Engineering Capstone Project Award

Graduation with Distinction

Soft Skills and Strengths

AdaptabilityProblem SolvingCollaboration

Team WorkingOpen-Mindedness

Critical ThinkingCommunication

Willingness to LearnResponsibilityReliability

EnthusiasmAutonomyMotivated

Stress ToleranceResearch and Analysing

Personal Time ManagementStrategic Planning

PatienceEmpathyDiscipline

Interests

- Deep Learning
 - Machine Learning
 - Cognitive Science
 - Agentic AI
 - Control Systems
 - Robotics
- Auto Chess
 - Card Games
 - Bees
 - Board Games
 - Billiards
 - Squash

September 2022 - February 2023

Research Assistant Toronto, On, Canada
Toronto Metropolitan University MEMS Lab

- Research on FPCB mirror-based lidar scanners
- Integration of PSD sensors, ToF sensors, and GUI

April 2021 - September 2021

Research Assistant Intern Toronto, On, Canada
Toronto Metropolitan University Ultrashort Laser Nanomanufacturing Lab

- Conducting porosity and fiber analyses via SEM image processing
- Generating molecular dynamics simulations

April 2020 - September 2020

Research Assistant Intern Toronto, On, Canada
Toronto Metropolitan University Ultrashort Laser Nanomanufacturing Lab

- Conducting porosity and fiber analyses via SEM image processing
- Generating molecular dynamics simulations

Publications

Manuscript 2025	Predicting the Risk of ACL Injuries Using Machine Learning Models and Tibial Anatomical Predictors , <i>Under Review</i> , Link Unavailable
Arxiv Preprint 2024	Morphological Detection and Classification of Microplastics and Nanoplastics Emerged from Consumer Products by Deep Learning , <i>IEEE</i> , Link
Journal Article 2023	MR. CAP: Multi-Robot Joint Control and Planning for Object Transport , <i>IEEE Control System Letters</i> , Link

Other Projects

- Time-series signal classification for tool and firearm use intervention via deep learning (Honda, GRYP Tech & UW)
- Data cleaning of wearable sensor time-series data (Honda, GRYP Tech & UW)
- IMU data processing and visualization (GRYP Tech & UW)
- Object segmentation and identification via SAM and YOLO (UW)
- Analyzing effect of homogeneous bagging on small medical datasets (UW)
- EKF design for 3D object tracking (UW)
- Improving SVM prediction accuracy on minority classes for imbalanced datasets (UW)
- MPC implementation for path planning and controls (TMU)
- Inverted pendulum control via PID controllers (TMU)

Software Skills

Data Analysis	PyTorch	TensorFlow
	AutoGluon	TabNet
	SkLearn	YOLO
	SAM	CATBoost
	XGBoost	Optuna
	NLopt	
Robotics	GTSAM	
Modeling and Simulation	Simulink	MATLAB
	ANSYS	Multisim